

C1902 刷入系统

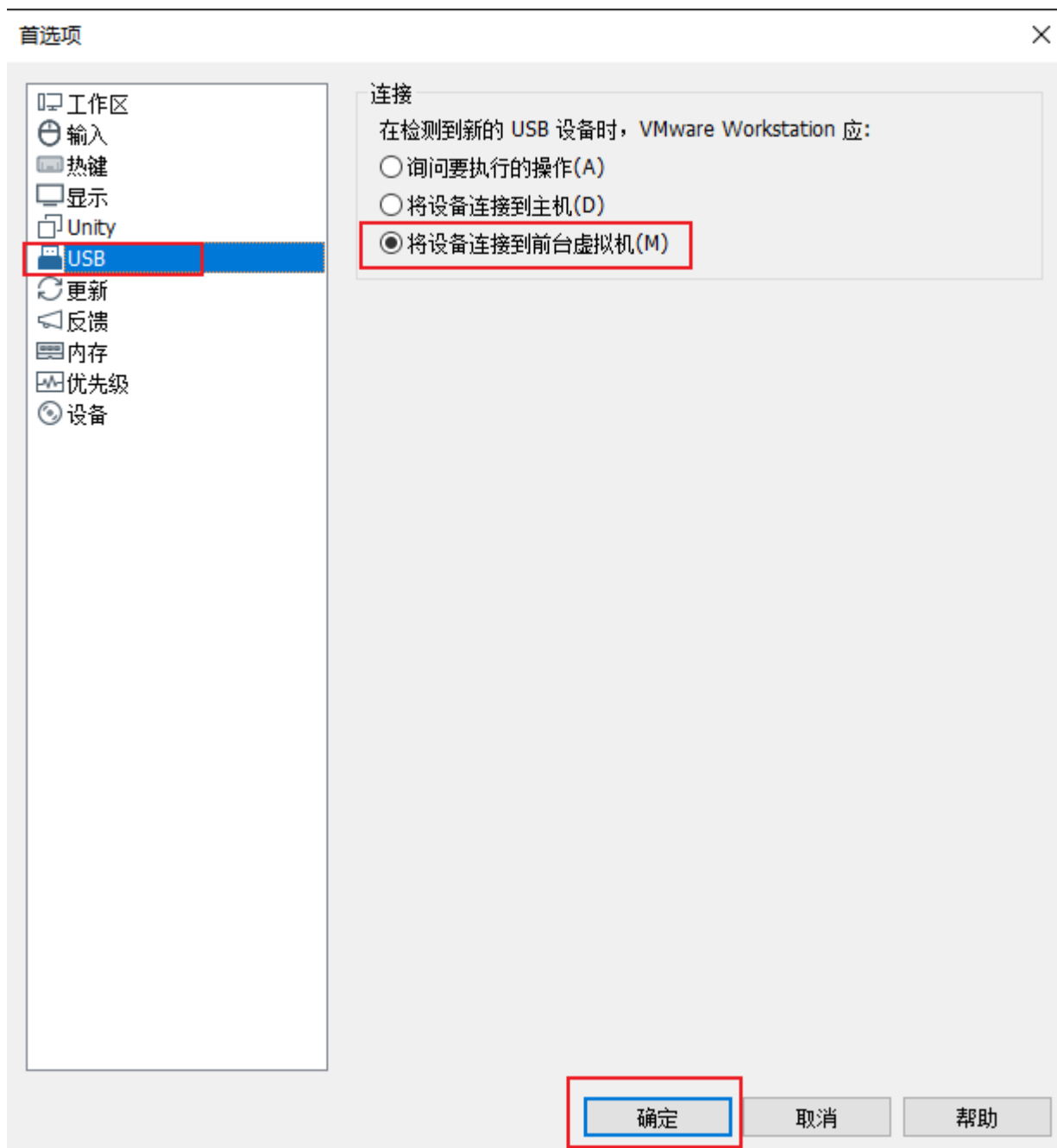
建议在 Linux 系统或 Linux 虚拟机下进行刷机操作，虚拟机和 SDK Manager 安装方法可参考 [VMware 虚拟机安装说明](#)。

1. 使用 NVIDIA SDK Manager 进行刷机

1.1 设置 VMware 的 USB 模式

将 USB 连接模式设置为「将设备连接到前台虚拟机」，避免刷机中途开发板重启导致失败。



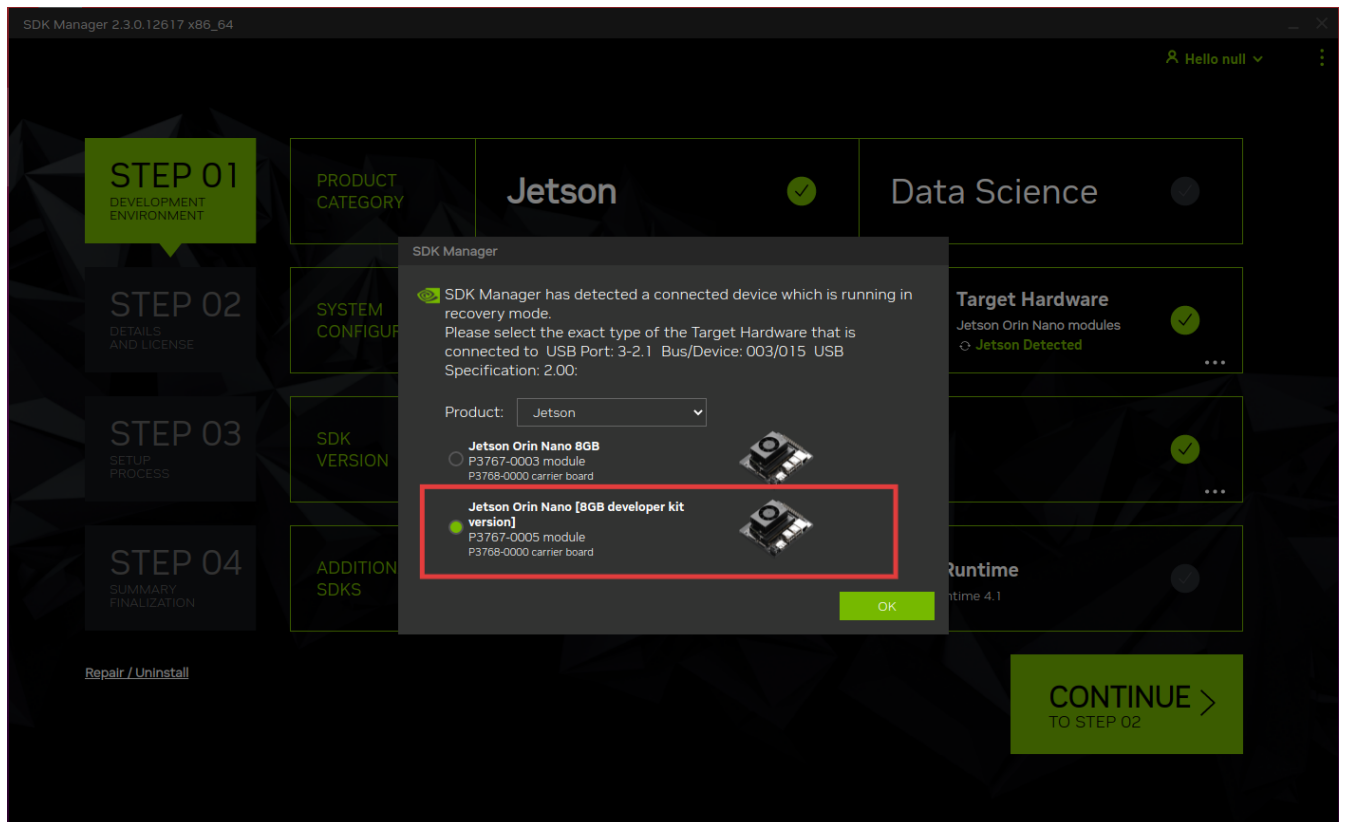


1.2 进入恢复模式

- 使用跳线帽短接开发板上的 **FC REC** 和 **GND** 针脚
- 使用 USB to Type-C 数据线将开发板连接到电脑
- 接通开发板的 DC 电源
- 成功进入恢复模式后，SDK Manager 会自动检测到核心卡类型，同时散热器风扇不会工作

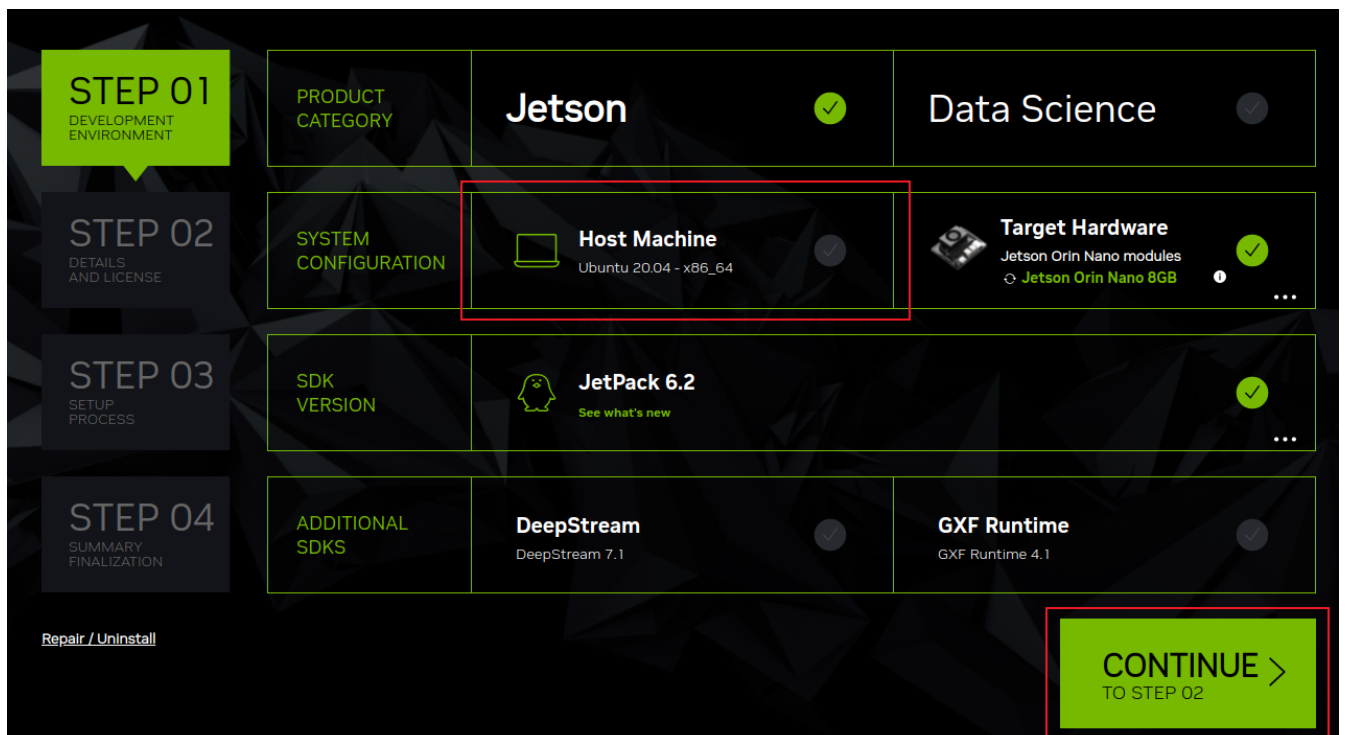
进入恢复模式后一段时间内不操作会重启进入系统，请注意操作时机。

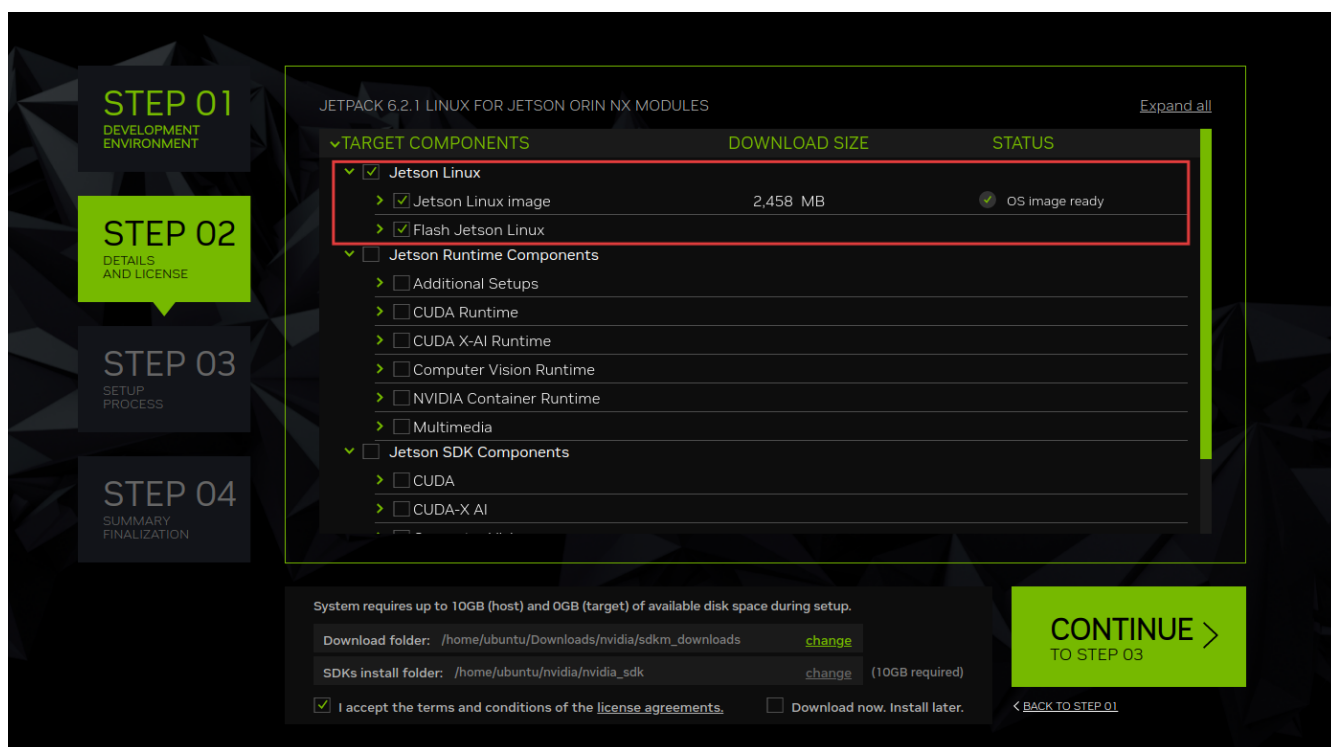




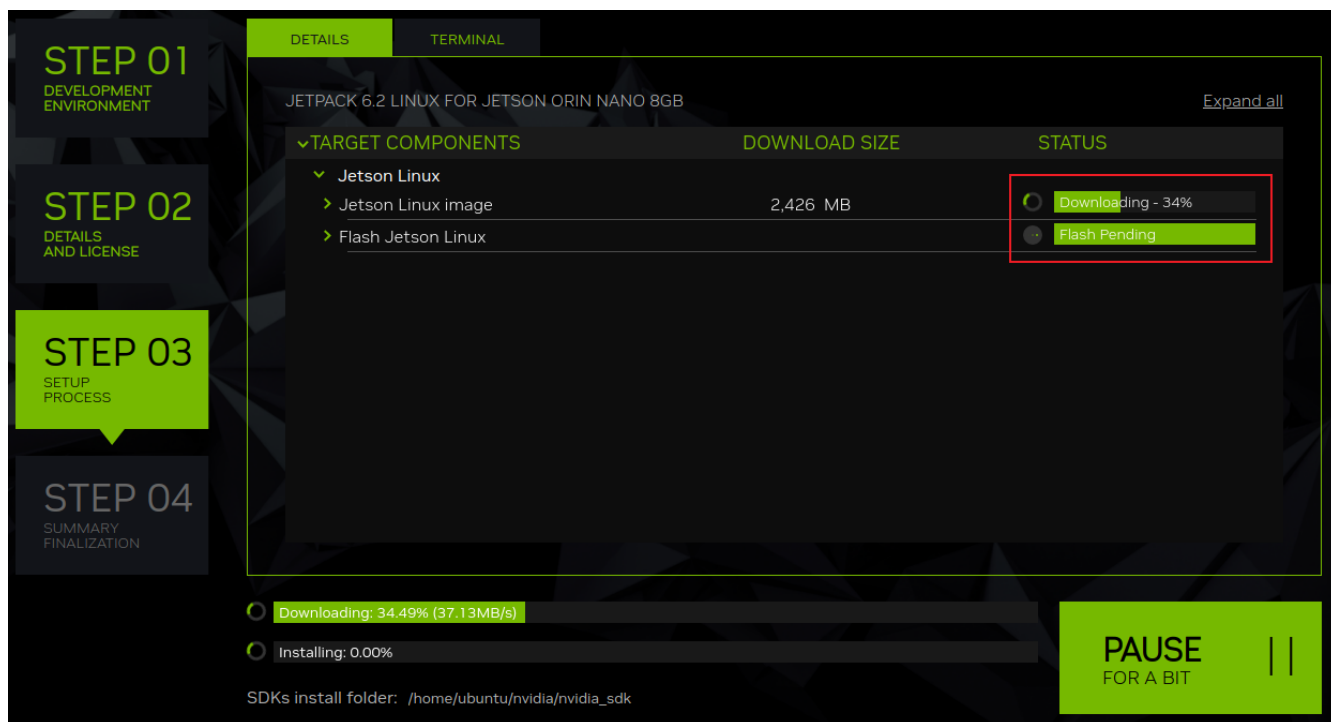
1.3 配置 SDK

- 取消 Host Machine，并点击 CONTINUE
- 只勾选 Jetson Linux





1.4 等待下载完成

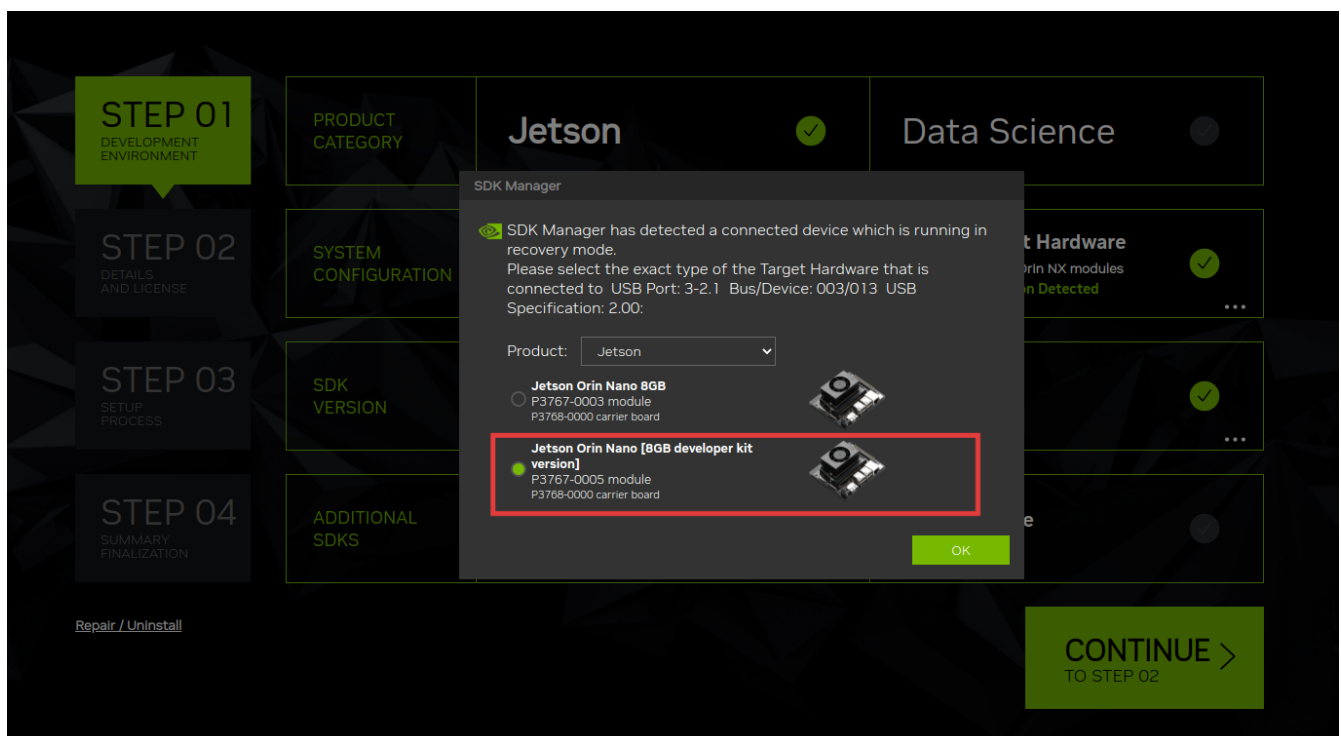
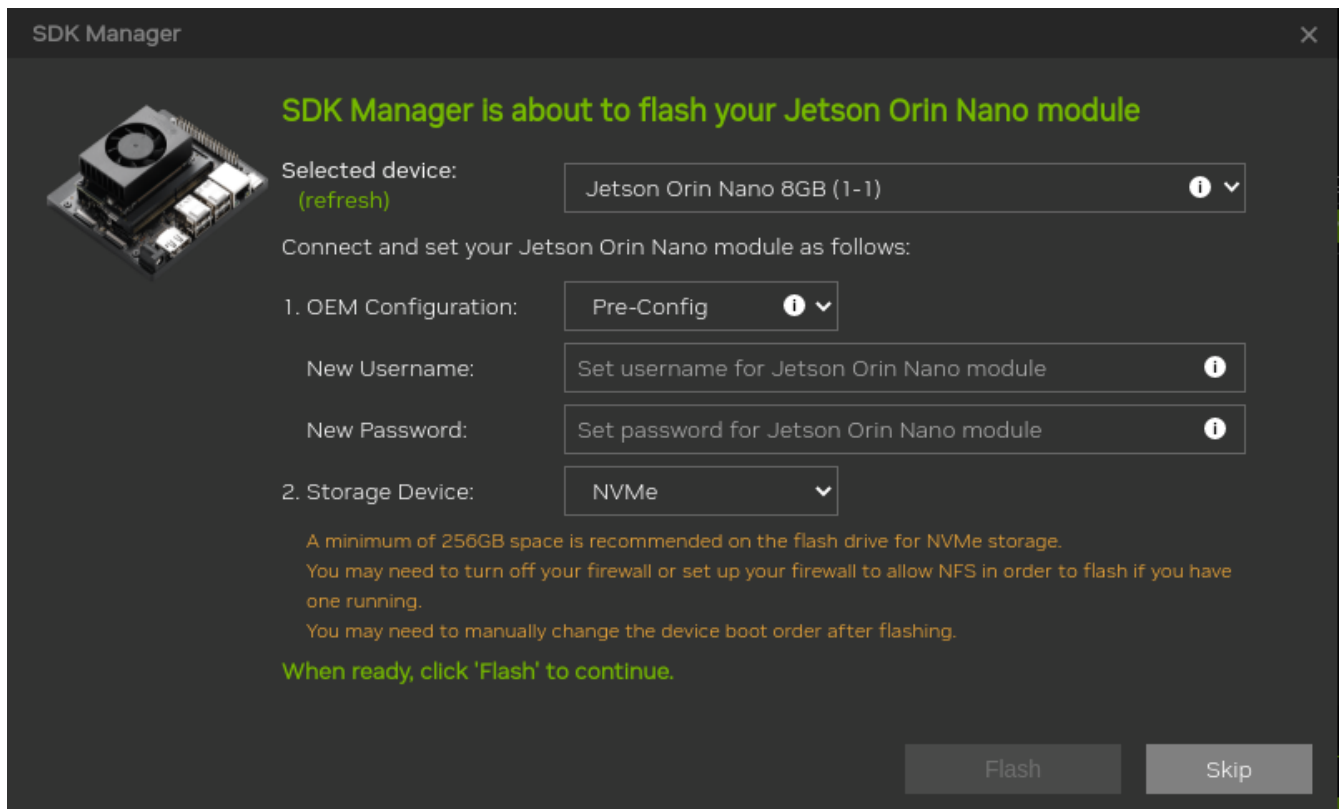


1.5 配置刷机参数

- **Pre-Config**: 预设账号密码
- **Runtime**: 开机后自行配置账号密码
- **Storage Device**: 系统刷入的介质（内存卡 / 固态硬盘 / U盘）
- 选择 **Developer Kit Version**

耐心等待刷机完成，大约 10~20 分钟。刷入过程中开发板可能会多次与主机重新连接，请勿途中拔开数据线或切断开发板电源。可以在识别到开发板类型以后就拔掉跳线帽。

刷入完成后，拔掉跳线帽，上电正常进入系统桌面或初始化设置页面即刷机成功。



STEP 01
DEVELOPMENT
ENVIRONMENT

STEP 02
DETAILS
AND LICENSE

STEP 03
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STEP 04
SUMMARY
FINALIZATION

DETAILS

TERMINAL

JETPACK 6.2 LINUX FOR JETSON ORIN NANO 8GB

Expand all

✓ TARGET COMPONENTS	DOWNLOAD SIZE	STATUS
✓ Jetson Linux		
➤ Jetson Linux image	2,426 MB	✓ OS image ready
➤ Flash Jetson Linux		✓ Success

✓ INSTALLATION COMPLETED SUCCESSFULLY.

EXPORT LOGS

FINISH
AND EXIT

✕

⏪ BACK TO STEP 01

SDK Manager 2.3.0.12617 x86_64

Hello null

STEP 01
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DETAILS

TERMINAL

Component Name: All

Message Type: All

Filter text here...

Find

✕

```
15:59:54 INFO: command finished successfully
16:01:04 DEBUG: running command < true >
16:01:04 INFO: command finished successfully
16:01:04 DEBUG: running command < lsusb | grep 0955: | awk '{printf $2 "/" $4 " " $6 ";"}' >
16:01:04 INFO: command finished successfully
16:01:10 DEBUG: running command < true >
16:01:10 INFO: command finished successfully
16:01:10 DEBUG: running command < lsusb | grep 0955: | awk '{printf $2 "/" $4 " " $6 ";"}' >
16:01:10 INFO: 001/021: 0955:7020;
16:01:10 INFO: command finished successfully
16:01:37 INFO: Flash Jetson Linux - flash: done.
16:01:37 INFO: Flash Jetson Linux - flash: Exiting with code 0
16:01:37 INFO: Flash Jetson Linux - flash: [ Component Install Finished Successfully ]
16:01:37 INFO: Flash Jetson Linux - flash: [host] [ 768.00 KB released. Disk Avail on Partition /dev/sda5: 198.96 GB ]
16:01:37 INFO: Flash Jetson Linux - flash: [ NV_L4T_FLASH_JETSON_LINUX_COMP Install took 17m53s ]
16:01:37 INFO: Flash Jetson Linux - flash: command finished successfully
16:01:37 SUMMARY: Flash Jetson Linux - flash: Install completed successfully.
16:01:37 INFO: All done!
```

✓ INSTALLATION COMPLETED SUCCESSFULLY.

EXPORT LOGS

FINISH
AND EXIT

✕

⏪ BACK TO STEP 01

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2. 使用命令行刷入 SUPER 固件

此操作需依赖官方固件环境，请先通过 SDK Manager 完成至少一次完整的烧录以创建固件缓存。

2.1 进入恢复模式

使用跳线帽短接 FC REC 和 GND 针脚，使用 Type-C 数据线连接至电脑并接通开发板电源。

2.2 使用官方固件进行命令行刷机

操作前请关闭正在运行的 SDK Manager 软件，刷入过程约为 10~20 分钟。

JetPack 6.2 版本刷入指令：

```
cd /home/ubuntu/nvidia/nvidia_sdk/JetPack_6.2.1_Linux_JETSON_ORIN_NANO_TARGETS/Linux_for_Tegra
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device nvme0n1p1 \
  -c tools/kernel_flash/flash_l4t_t234_nvme.xml -p "-c bootloader/generic/cfg/flash_t234_qspi.xml" \
  --showlogs --network usb0 jetson-orin-nano-devkit-super internal
```

JetPack 5.1.5 版本刷入指令：

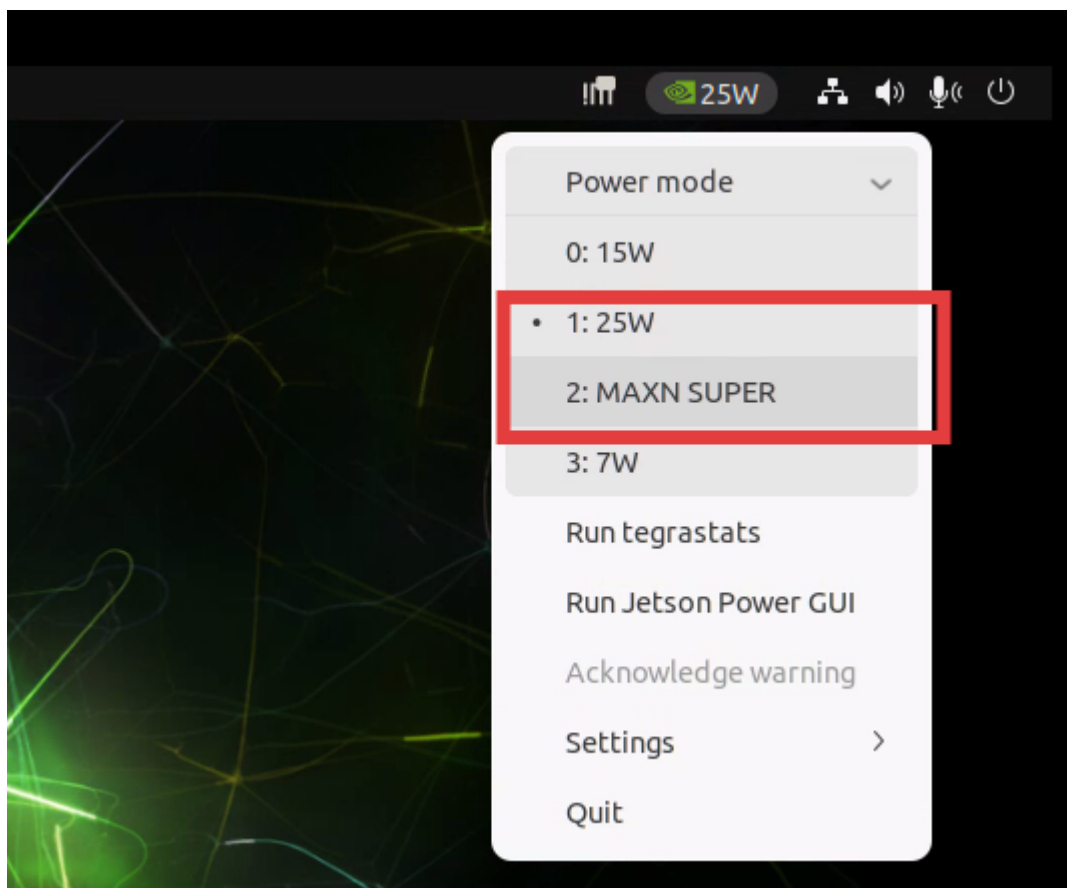

```
cd /home/ubuntu/nvidia/nvidia_sdk/JetPack_5.1.5_Linux_JETSON_ORIN_NANO_TARGETS/Linux_for_Tegra
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device nvme0n1p1 \
-c tools/kernel_flash/flash_l4t_external.xml -p "-c bootloader/t186ref/cfg/flash_t234_qspi.xml" \
--showlogs --network usb0 jetson-orin-nano-devkit-super internal
```



```
ubuntu@ubuntu-virtual-machine: ~/nvidia/nvidia_sdk/JetPac...
00:03:33.134 - Debug: Sha1 checksum matched for /mnt/internal/qspi_bootblob_ver.txt
00:03:33.137 - Debug: Writing /mnt/internal/qspi_bootblob_ver.txt (109 bytes) in
to /dev/mtd0:66977792
Copied 109 bytes from /mnt/internal/qspi_bootblob_ver.txt to address 0x03fe0000
in flash
00:03:33.164 - Debug: Writing gpt_secondary_3_0.bin (parittion: secondary_gpt) i
nto /dev/mtd0
00:03:33.182 - Debug: Sha1 checksum matched for /mnt/internal/gpt_secondary_3_0.
bin
00:03:33.185 - Debug: Writing /mnt/internal/gpt_secondary_3_0.bin (16896 bytes)
into /dev/mtd0:67091968
Copied 16896 bytes from /mnt/internal/gpt_secondary_3_0.bin to address 0x03ffbe0
0 in flash
00:03:33.219 - Info: Successfully flashed the QSPI.
00:03:33.222 - Info: Flashing success
00:03:33.244 - Debug: The device size indicated in the partition layout xml is s
maller than the actual size. This utility will try to fix the GPT.
Flash is successful
Reboot device
Cleaning up...
Log is saved to Linux_for_Tegra/initrdlog/flash_1-1_0_20250709-164603.log
ubuntu@ubuntu-virtual-machine:~/nvidia/nvidia_sdk/JetPack_6.2.1_Linux_JETSON_ORI
N_NANO_TARGETS/Linux_for_Tegra$
```

2.3 验证是否处于 SUPER 模式

开发板上电开机完成用户配置后，在桌面右上角可以选择电源模式。**25W & MAXN SUPER** 为 SUPER 模式独有，普通模式只有 7W 和 15W 两档。



3. 备份与恢复现有固件

3.1 备份固件

使用跳线帽短接 FC REC 和 GND 针脚，Type-C 连接电脑，接通电源进入恢复模式。

或在正常开机状态下连接到主机，输入以下命令重启到恢复模式：

```
sudo reboot -f forced-recovery
```

进入原刷机固件缓存下的命令行目录：

```
cd /home/ubuntu/nvidia/nvidia_sdk/JetPack_6.2.1_Linux_JETSON_ORIN_NANO_TARGETS/Linux_for_Tegra
```

安装依赖包：

```
sudo apt-get install qemu-user-static libxml2-utils abootimg sshpass nfs-kernel-server binutils
```

备份命令：

```
sudo ./tools/backup_restore/l4t_backup_restore.sh -b -e nvme0n1 jetson-orin-nano-devkit-nvme
```

恢复命令：

```
sudo ./tools/backup_restore/l4t_backup_restore.sh -r -e nvme0n1 jetson-orin-nano-devkit-nvme
```

若使用第三方载板恢复备份，需要修改 `nvrestore_partitions.sh` 注释掉 292~296 行的内容。

```
278 # This block will check to make sure the model of the target board matches the
279 # model of the board the images came from.
280 for value in $(grep -v -e '(\^ *$|^#)' < "${FILE_NAME}"); do
281     declare -a FIELDS
282     for part in {1..6}; do
283         FIELDS[part]=$(echo "$value" | awk -F, -v part=${part} '{print $part}')
284     done
285     if [ "${FIELDS[1]}" = 'board_spec' ]; then
286         set_board_spec
287         if [[ "${FIELDS[2]}" == "${BOARD_SPEC}" ]]; then
288             BOARD_MATCH=true
289         fi
290     fi
291 done
292 if [ ${BOARD_MATCH} = false ]; then
293 #     echo "${SCRIPT_NAME}: You are trying to flash images from a board model that does not"
294 #     echo "match the current board you're flashing onto."
295 #     exit 1
296 #fi
297
298 declare -A able_to_delete
```